

# Player movement at low speeds is biased towards cardinal directions

MC-184530 PLAYER

|                 |                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------|
| Status          | Resolved / Fixed                                                                             |
| Affected        | 1.15.2; 20w20b; 20w21a; 1.17.1; 1.18.2; 22w15a; 23w51b; 1.21.3; 1.21.4 Pre-Release 1; 1.21.4 |
| Fixed in        | 25w02a                                                                                       |
| Reported        | 2020-05-19                                                                                   |
| Mojira category | Player                                                                                       |

## REPRODUCTION TARGET

Use version: 1.15.2

Affected versions (10):

1.15.2

20w20b

20w21a

1.17.1

1.18.2

22w15a

23w51b

1.21.3

1.21.4 Pre-Release 1

1.21.4

Environment:

Minecraft Java Edition 1.15.2 in a single-player, default Superflat world. Use Creative mode with cheats enabled so command blocks can be created and edited. Use only one player because the report's repeating command targets every player with @a. The deterministic layout below places an impulse command block at X=0, Y=4, Z=0, a repeating command block at X=1, Y=4, Z=0, and the starting point south of them at X=0.5, Y=4, Z=10.5. Both command blocks remain Unconditional and Needs Redstone. The lever-on-impulse arrangement and command-block input behavior were operationally checked against Minecraft Wiki documentation. ([minecraft.wiki](https://minecraft.wiki/w/Tutorial%3AQuasi-connectivity?utm\_source=openai))

## PRECONDITIONS

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- Launch the affected Java Edition version 1.15.2, which is the first version listed by the report. Do not use fix version 25w02a.
- Create a new single-player world using the default Superflat preset, Creative mode, and Allow Cheats: ON.
- Use a world with only one player connected so the @a selector produces one armor-stand trail rather than trails for multiple players.
- Ensure the Move Forward and Sneak actions are available. With default controls these are W and Left Shift.
- Do not move the mouse during either measured walk. Camera yaw is set by the teleport command, and manual camera movement would invalidate the comparison.
- Keep the lever OFF until the first measured run begins.

## STEPS TO REPRODUCE

### 1 **REPORT**

From the Minecraft Launcher, start an installation using Java Edition 1.15.2 and wait for the main menu.

**INPUT** Select or create a launcher installation for release 1.15.2, then click Play.

**SUCCESS CUE** The lower-left corner of the Minecraft main menu identifies the running version as Minecraft 1.15.2.

### 2 **INFERRED**

Create the test world as a Creative, cheat-enabled Superflat world.

**INPUT**

Singleplayer -> Create New World -> Game Mode: Creative -> More World Options... -> Allow Cheats: ON -> cycle World Type to Superflat -> Done -> Create New World

**SUCCESS CUE** The player loads into a flat grass world in Creative mode and can execute commands.

### 3 **INFERRED**

Wait for the world to become ready before constructing the apparatus.

**SUCCESS CUE** Nearby terrain has rendered, the player is standing on the flat surface, and movement and chat commands respond normally.

### 4 **INFERRED**

Move the player to the deterministic construction area near the world origin.

**INPUT** Press T, enter `/tp @s 0.5 4 4.5 180 0``, and press Enter.

**SUCCESS CUE** The player is standing on the grass surface near X=0, Z=4.5 and is facing approximately north toward decreasing Z.

### 5 **INFERRED**

Place the impulse command block at the defined test origin using a chat command.

**INPUT** Press T, enter `/setblock 0 4 0 minecraft:command_block``, and press Enter.

**SUCCESS CUE** An orange impulse command block is visible on the surface at X=0, Y=4, Z=0.

### 6 **INFERRED**

Place the repeating command block directly east of and adjacent to the impulse command block.

**INPUT** Press T, enter `/setblock 1 4 0 minecraft:repeating_command_block``, and press Enter.

**SUCCESS CUE** A purple repeating command block is visible at X=1, Y=4, Z=0, sharing a side with the impulse command block.

## 7 INFERRED

Configure the impulse command block to teleport the nearest player to a point 10 blocks south of it with yaw 174 and pitch 0. Yaw 174 is six degrees away from north-facing yaw 180, as required by the report's south-of-the-blocks example.

### INPUT

Right-click the impulse command block at 0 4 0. In its command field enter `/tp @p 0.5 4 1 0.5 174 0``. Verify the buttons read Impulse, Unconditional, and Needs Redstone, then click Done.

**SUCCESS CUE** The impulse command block has saved the teleport command and remains configured as Impulse, Unconditional, Needs Redstone.

## 8 REPORT

Configure the repeating command block with the report's exact armor-stand trail command.

### INPUT

Right-click the repeating command block at 1 4 0. Enter `/execute as @a as @s run summon minecraft:armor_stand ~ ~ ~ {Small:1b}`` exactly, including `{Small:1b}``. Verify the buttons read Repeat, Unconditional, and Needs Redstone, then click Done.

**SUCCESS CUE** The repeating command block has saved the exact report command and remains configured as Repeat, Unconditional, Needs Redstone.

## 9 INFERRED

Obtain one lever for controlling both adjacent command blocks.

**INPUT** Press T, enter `/give @s minecraft:lever 1``, and press Enter.

**SUCCESS CUE** A lever is present in the player's inventory or hotbar.

## 10 INFERRED

Place the lever on the top face of the impulse command block without opening the command-block interface. Sneaking while placing allows the lever to be placed on an interactive block.

### INPUT

Select the lever in the hotbar. Hold Left Shift, right-click the top face of the impulse command block at 0 4 0, then release Left Shift.

**SUCCESS CUE** An OFF lever is attached to the top of the impulse command block at approximately X=0, Y=5, Z=0. No teleport has occurred and no armor stands are spawning.

## 11 VERIFIED

Confirm the final command-block power configuration before beginning the measured runs.

### INPUT

Inspect each command block by right-clicking its exposed side if necessary. Confirm the impulse block is Impulse/Unconditional/Needs Redstone and the repeating block is Repeat/Unconditional/Needs Redstone. Close each interface with Done without changing the commands.

**SUCCESS CUE** Both commands remain saved, both command blocks require redstone power, and the lever is still OFF.

## 12 **INFERRED**

Position the player immediately south of the apparatus so the lever can be targeted before the teleport fires.

**INPUT** Press T, enter `/tp @s 0.5 4 2.5 180 0`, and press Enter.

**SUCCESS CUE** The player stands about two blocks south of the impulse command block, on the ground and within interaction reach of the lever.

## 13 **REPORT**

Aim the crosshair directly at the lever, stop all movement, and then stop providing mouse movement.

**INPUT**

Move the mouse until the lever is outlined under the crosshair. Release all movement keys, then physically stop moving the mouse or otherwise provide zero camera input.

**SUCCESS CUE** The lever is under the crosshair, the player is stationary, and the lever remains OFF.

## 14 **REPORT**

Activate the lever to power both command blocks and initiate the normal-speed control run. Do not move the mouse after clicking.

**INPUT** Right-click once.

**SUCCESS CUE** The lever's rising signal causes the player to arrive at X=0.5, Y=4, Z=10.5 with yaw 174 and pitch 0; the player is standing on the flat ground and the repeating command block is powered.

## 15 **REPORT**

Perform the normal-speed control walk without sneaking and without changing the camera angle.

**INPUT** Press and hold W. Do not press Left Shift and do not move the mouse.

**SUCCESS CUE** The player moves forward from the teleported starting position toward the command-block apparatus at normal walking speed while small armor stands are left along the traversed route.

## 16 **REPORT**

Continue the normal-speed walk only until the player is close enough to reach the lever, then stop before adjusting the view.

**INPUT**

Continue holding W until the command blocks are approximately 2 to 4 blocks ahead, then release W.

**SUCCESS CUE** The player is stationary near the apparatus and within right-click reach of the lever; the first armor-stand trail remains in the world.

## 17 **REPORT**

After forward movement has completely stopped, aim at the elevated lever and switch it OFF to stop the repeating command block.

**INPUT** Move the mouse to place the crosshair on the lever, then right-click once.

**SUCCESS CUE** The lever is visibly OFF, the player is not teleported by the falling signal, and new armor stands stop appearing.

## 18 **REPORT**

Prepare the second run from the same nearby position. Aim at the lever again, ensure neither W nor Left Shift is held, and stop moving the mouse.

### INPUT

Place the crosshair on the lever, release all movement and sneak inputs, then provide no further mouse movement.

**SUCCESS CUE** The stationary player is within reach of the OFF lever and has the lever under the crosshair.

## 19 **REPORT**

Activate the lever a second time to reset the player to the identical starting coordinates and camera rotation for the slow, sneaking run.

**INPUT** Right-click once.

**SUCCESS CUE** The player is again teleported to X=0.5, Y=4, Z=10.5 with yaw 174 and pitch 0, and the repeating command block is powered again.

## 20 **REPORT** **TRIGGER**

Begin the low-speed test movement by sneaking and moving forward while keeping the camera completely unchanged.

### INPUT

Press and hold Left Shift, then press and hold W while continuing to hold Left Shift. Do not move the mouse.

**SUCCESS CUE** The player is visibly in the lowered sneaking posture and is moving forward from the same starting location toward the command blocks.

## 21 **REPORT**

Continue holding both movement inputs until the player is again close enough to operate the lever.

### INPUT

Keep holding W and Left Shift, with no mouse movement, until the command blocks are approximately 2 to 4 blocks ahead.

**SUCCESS CUE** The player has traversed the test route while continuously sneaking and is within interaction reach of the lever; both runs' armor stands remain present.

## 22 **REPORT**

Stop the sneaking movement before changing the camera direction.

### INPUT

Release W, then release Left Shift. Keep the mouse stationary until both inputs are released.

**SUCCESS CUE** The player is stationary near the apparatus and is no longer moving or sneaking.

## 23 **REPORT**

Aim at the elevated lever after the measured movement has ended.

### INPUT

Move the mouse until the crosshair outlines the lever on top of the impulse command block.

**SUCCESS CUE** The powered lever is under the crosshair and the player remains stationary.

## 24 **REPORT**

Switch the lever OFF once to stop further armor-stand summoning, then make no further movement through the recorded paths.

**INPUT** Right-click once.

**SUCCESS CUE** The lever is OFF, the repeating command block is no longer summoning armor stands, the player remains near the apparatus, and the two completed armor-stand trails are available for the observation subsystem.

## HANDOFF TO OBSERVATION

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The normal-speed control run and the sneaking run have both been completed from the same teleport position and yaw. W and Left Shift are released, the player is stationary near the command blocks, the lever has just been switched OFF, the repeating command block has stopped summoning armor stands, and both recorded trails remain in place. Hand control to the observation subsystem to inspect the trails without moving through or removing them.

## UNCERTAINTIES

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- The report does not specify fixed world coordinates, so the command blocks at 0 4 0 and 1 4 0 and the starting point at 0.5 4 10.5 are deterministic practical choices that preserve the stated geometry: a flat plane, adjacent command blocks, and a starting point about 10 blocks directly south of them.
- The report does not supply the complete teleport command or target selector. `/tp @p 0.5 4 10.5 174 0`` is constructed from the report's requirement for a position south of the command blocks and yaw 174, six degrees away from north-facing yaw 180.
- The report does not explicitly specify a game mode. Creative mode with cheats is selected because Java Edition command blocks must be created and edited with the necessary permissions; this is an execution requirement rather than a claimed bug condition.
- The instruction to stop approximately 2 to 4 blocks from the apparatus operationalizes the report's phrase 'until you are close to the lever.' The exact endpoint is not asserted to affect reproduction as long as the same forward input and fixed yaw are maintained until the player is within interaction reach.
- The report says the lever can be placed directly on either command block. This plan chooses the top of the impulse block so activating it gives the impulse block a rising signal and powers the adjacent repeating block through the powered arrangement.

MANUAL RESULT

|                  |                    |
|------------------|--------------------|
| Version used     |                    |
| Reproduced?      | YES / NO / PARTIAL |
| Trigger reached? | YES / NO           |
| Crash file       |                    |
| Notes            |                    |
| Tested by / date |                    |